14.6.2.3.3 Python getopt module

The getopt module of Python helps you to parse (split) command-line options and arguments. This module provides getopt() method to enable command-line argument parsing.

getopt.getopt function

This function parses command-line options and parameter list. Following is the syntax for this method –

<opts>,<args>=getopt.getopt(argv, options, [long_options])

Here is the detail of the parameters –

argv – This is the argument list of values to be parsed (splited). In our program the complete command will be passed as a list. For example ***c:\pyprg\pali.py -i c:\pyprg\pali_cpp***

options – This is string of option letters that the Python program recognize as, for input or for output, with options (like 'i' or 'o') that followed by a colon (:). Here colon is used to denote the mode.

long_options – This contains a list of strings. Argument of Long options should be followed by an equal sign '='. In our program the C++ file name along with its path will be passed as string and 'i' i will be also passed to indicate it as the input file.

getopt() method returns value consisting of two elements. Each of these values are stored separately in two different list (arrays) **opts** and **args** .Opts contains list of splitted strings like mode and path. **args** contains error string, if at all the comment is given with wrong path or mode. **args** will be an empty list if there is no error.

For example The Python code which is going to execute the **C++ file p4** in command line will have the getopt() method like the following one.

opts, args = getopt.getopt (argv, "i:",['ifile='])

where **opts** contains	[('-i', 'c:\\pyprg\\p4')]
-i :-	**option - mode should be followed by : (colon)**
'c:\\pyprg\\p4'	**value - absolute path of C++ file.**

In our examples since the entire command line commands are parsed and no leftover argument, the **second argument args** will be empty []. If args is displayed using print() command it displays the output as [].

```
>>>print(args)
[]
```


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